

Approximate calculations with the derivative

Near one point a curve is a straight line — which is enough to compute $\sqrt{4.02}$ in your head.

LESSON 28–30

3 × 40 MIN

ALGEBRA 11, §2.1

P1 · 9.4 · P2 · NUMERICAL METHODS

LESSON CLOCK

14'	21'	21'	21'	33'	10
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Where the formula comes from 14 min

Roots and powers 21 min

Changes in a quantity 21 min

Error estimates 21 min

Practice 33 min

Homework 10 min

LEARNING OBJECTIVES

- Derive and use the linear approximation $f(a + h) \approx f(a) + h \cdot f'(a)$.
- Estimate a root or a power without a calculator.
- Estimate the change in a quantity from a small change in another.
- Estimate the relative and percentage error in a computed quantity.

TEXTBOOK

National pp. 131–146

Cambridge Pure Mathematics
1, pp. 186–190

Workbook P1 Exercise 9C

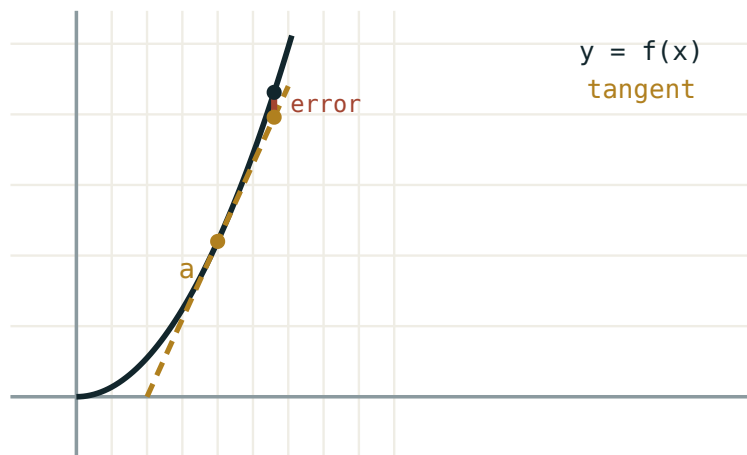
01 Explanation

Where the formula comes from

Return to the definition. For small h ,

$$\frac{f(a + h) - f(a)}{h} \approx f'(a) \Rightarrow f(a + h) \approx f(a) + h \cdot f'(a)$$

Geometrically: near the point of tangency the curve and its tangent are almost the same line, so the tangent's height is a good estimate of the curve's.



near a the tangent and the curve agree

Close to a the two agree; the error grows with the square of the distance.

CHOOSING A

Pick the nearest point at which f and f' are **easy**. For $\sqrt{4.02}$ take $a = 4$; for $\sin 31^\circ$ take $a = 30^\circ$. The whole art is that choice.

Roots and powers

Estimate $\sqrt{4.02}$. Take $f(x) = \sqrt{x}$, $a = 4$, $h = 0.02$:

$$f(4) = 2, f'(x) = \frac{1}{2\sqrt{x}}, f'(4) = 0.25$$

$$\sqrt{4.02} \approx 2 + 0.02 \times 0.25 = 2.005$$

The true value is $2.004994\dots$ — the estimate is right to five decimal places, from two multiplications.

ESTIMATE	A	H	ANSWER
$\sqrt{9.06}$	9	0.06	3.01
$\sqrt{101}$	100	1	10.05
2.01^3	2	0.01	8.12
$\frac{1}{4.98}$	5	-0.02	0.2008

Changes in a quantity

The same formula, read as a statement about changes:

$$\Delta y \approx f'(x) \cdot \Delta x$$

Example. A cube's side is measured as 10 cm, with a possible error of 0.05 cm. By how much can the volume be out?

$$V = a^3, \frac{dV}{da} = 3a^2 = 300, \Delta V \approx 300 \times 0.05 = 15 \text{ cm}^3$$

So $V = 1000 \pm 15 \text{ cm}^3$. The absolute error in the side is small, but it is multiplied by $3a^2$ on its way to the volume.

Relative and percentage error

$$\text{relative error } \frac{\Delta y}{y} \approx \frac{f'(x) \cdot \Delta x}{f(x)}$$

For the cube: $\frac{15}{1000} = 1.5\%$, from a side error of only $\frac{0.05}{10} = 0.5\%$. The relative error is tripled, because the volume is the **third** power of the length.

A RULE WORTH REMEMBERING

For $y = x^n$ the relative errors satisfy $\frac{\Delta y}{y} \approx n \cdot \frac{\Delta x}{x}$. Squaring doubles the percentage error; cubing triples it; taking a square root halves it.

THE APPROXIMATION IS ONE-SIDED IN ACCURACY

It is good for **small** h only. Estimating $\sqrt{4.9}$ from $a = 4$ gives 2.1125 against a true 2.2136 — an error of 5%. Choose the nearer convenient point.

02 Worked examples

EXAMPLE 1 Estimate $\sqrt{26}$ without a calculator.

① $a = 25, h = 1$
 $f(25) = 5$

② $f(x) = \frac{1}{2\sqrt{x}}, f(25) = 0.1$

③ $\sqrt{26} \approx 5 + 1 \times 0.1 = 5.1$
 True: 5.0990.

ANSWER

≈ 5.1

EXAMPLE 2 Estimate 3.02^4 .

① $f(x) = x^4, a = 3, h = 0.02$
 $f(3) = 81$

② $f(x) = 4x^3, f(3) = 108$

③ $81 + 0.02 \times 108 = 83.16$
 True: 83.17.

ANSWER

≈ 83.16

EXAMPLE 3 A sphere's radius is 6 ± 0.1 cm. Estimate the error in its volume.

$$1 \quad V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2 = 144\pi$$

$$2 \quad \Delta V \approx 144\pi \times 0.1 \approx 45.2$$

$$3 \quad V = 288\pi \approx 904.8$$

$$4 \quad \text{Relative error } \frac{45.2}{904.8} = 5\%$$

$$\text{Three times the } \frac{0.1}{6} \approx 1.67\% \text{ in } r.$$

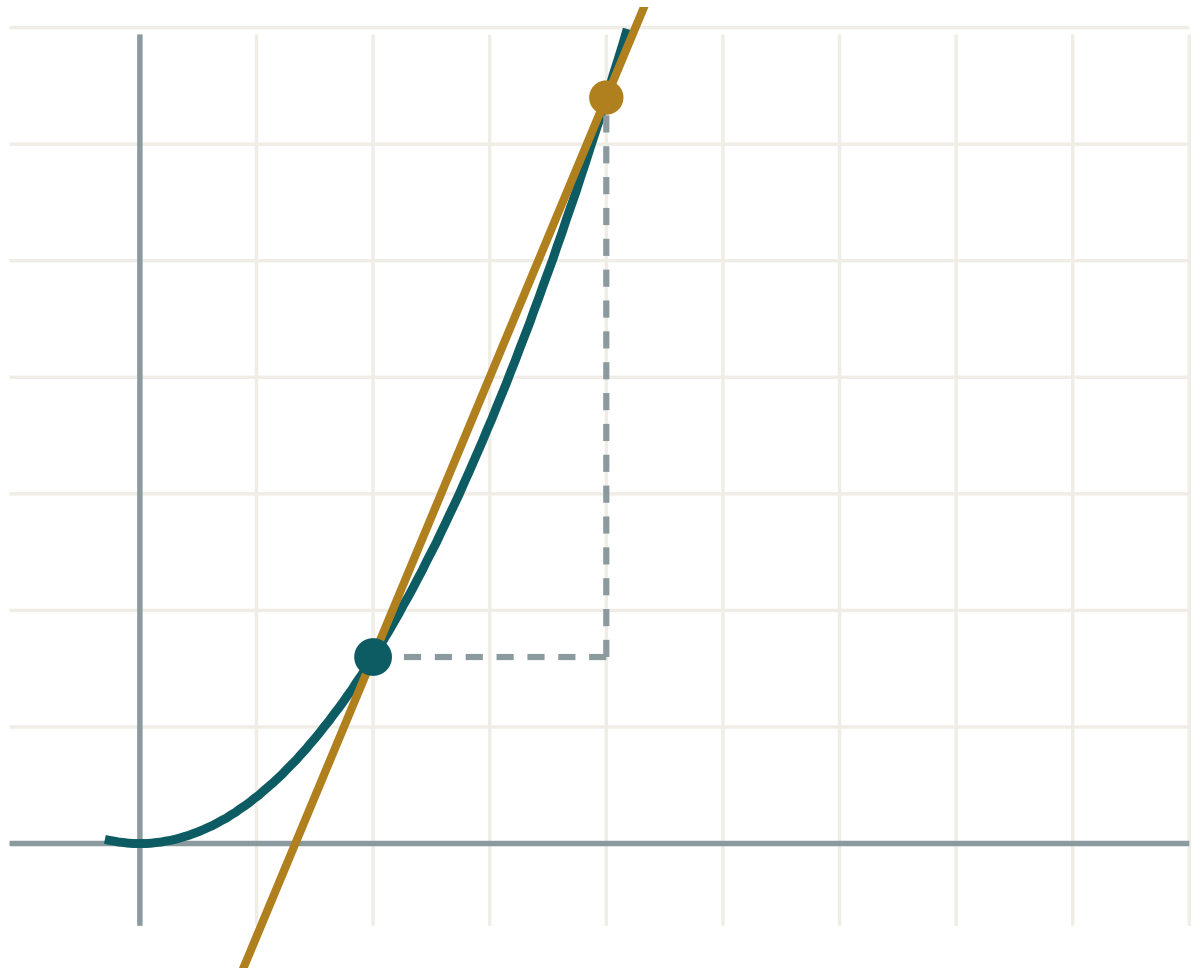
ANSWER

$$\Delta V \approx 45 \text{ cm}^3, \text{ about } 5\%$$

03 Interactive model

Shrink h and watch the tangent's prediction converge on the curve's value.

The tangent as a predictor



x 2

h 2

 $\Delta y / \Delta x$ 2.4 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Approximation	Taqribiy hisoblash	Приближённое вычисление
Linear approximation	Chiziqli yaqinlashish	Линейное приближение
Small increment	Kichik ortirma	Малое приращение
Differential	Differensial	Дифференциал
Absolute error	Absolyut xatolik	Абсолютная погрешность
Relative error	Nisbiy xatolik	Относительная погрешность
Percentage error	Foizli xatolik	Процентная погрешность
Convenient point	Qulay nuqta	Удобная точка
Order of accuracy	Aniqlik tartibi	Порядок точности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The linear approximation is:

$$f(a) \cdot h$$

$$f(a) + h f'(a)$$

$$f'(a) + h$$

$$f(a + h)$$

For $\sqrt{4.02}$ the best a is:

$$0$$

$$1$$

$$4$$

$$5$$

$\Delta y \approx$:

$$f'(x) \Delta x$$

$$f(x) \Delta x$$

$$\Delta x$$

$$f'(x)$$

For $y = x^3$ a 1% error in x gives about:

For $y = \sqrt{x}$ a 4% error in x gives about:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Estimate $\sqrt{4.04}$

ANSWER ≈ 2.01

2. Estimate $\sqrt{9.06}$

ANSWER ≈ 3.01

3. Estimate $\sqrt{16.08}$

ANSWER ≈ 4.01

4. Estimate 2.01^2

ANSWER ≈ 4.04

5. Estimate 3.02^2

ANSWER ≈ 9.12

6. Estimate 1.02^3

ANSWER ≈ 1.06

7. Δy for $y = 5x$ when $\Delta x = 0.2$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. Estimate $\sqrt{26}$

ANSWER ≈ 5.1

2. Estimate $\sqrt{101}$

ANSWER ≈ 10.05

3. Estimate 3.02^4

ANSWER ≈ 83.16

4. Estimate $\frac{1}{4.98}$

ANSWER ≈ 0.2008

5. Cube side 10 ± 0.05 ; error in the volume

ANSWER $\approx 15 \text{ cm}^3$

6. Sphere radius 6 ± 0.1 ; error in the volume

ANSWER $\approx 45 \text{ cm}^3$

7. A 2% error in the radius gives what error in the area of a circle?

ANSWER $\approx 4\%$

Hard · stretch and reason

7 PROBLEMS

1. Estimate $\sqrt[3]{8.12}$

ANSWER ≈ 2.01

2. Estimate $\sqrt{48}$ from $a = 49$

ANSWER ≈ 6.929 (true 6.928)

3. A circle's radius grows 1%. By what percentage does the area grow?

ANSWER $\approx 2\%$

4. A cube's surface area is measured to 2%. To what accuracy is the volume known?

ANSWER $\approx 3\%$

5. Estimate the error in $T = 2\pi\sqrt{\frac{L}{g}}$ from a 1% error in L

ANSWER $\approx 0.5\%$

6. Show the error in the linear approximation is of order h^2

ANSWER From the next term of the expansion

7. Why does estimating $\sqrt{4.9}$ from $a = 4$ fail badly?

ANSWER $h = 0.9$ is not small — the h^2 term dominates

07 Homework

Homework — 6 tasks

State a , h and $f'(a)$ before every estimate.

1. Estimate $\sqrt{36.6}$ and $\sqrt{99}$.

2. Estimate 2.03^5 .

3. Estimate $\frac{1}{2.98}$.

4. A cube's side is 8 ± 0.02 cm. Find the volume and its absolute and percentage error.

5. A sphere's radius is measured to within 1%. To what accuracy is its volume known?
6. Explain in three sentences why the approximation is only good for small h .

Modelling with the derivative

Four lessons of real situations — motion, growth, cost, flow — in which the derivative is the quantity the question is really about.

LESSON 31–34

4 × 40 MIN

ALGEBRA 11, §2.2

P1 · 9.4

LESSON CLOCK

14

27'

27'

27'

44'

21'

The derivative has units	14 min
Motion	27 min
Connected rates	27 min
Cost and revenue	27 min
Practice	44 min
Homework and consolidation	21 min

LEARNING OBJECTIVES

- Interpret a derivative as a rate of change in a stated context, with units.
- Model motion with displacement, velocity and acceleration.
- Use connected rates of change.
- Interpret marginal cost and marginal revenue.

TEXTBOOK

National	pp. 147–168
Cambridge	Pure Mathematics 1, pp. 186–190
Workbook	P1 Exercise 9C

01

Explanation

The derivative has units

READ THE UNITS OFF THE FRACTION

$\frac{dy}{dx}$ carries the units of y divided by those of x . A distance in metres over a time in seconds gives m/s; a cost in so'm over a number of items gives so'm per item. Stating the units is half the interpretation.

Y	X	$\frac{DY}{DX}$	UNITS
distance	time	velocity	m/s
velocity	time	acceleration	m/s^2
volume	time	flow rate	cm^3/s
total cost	quantity	marginal cost	so'm per item
population	time	growth rate	people per year

Motion

$$v(t) = s'(t) \quad a(t) = v'(t) = s''(t)$$

QUESTION IN WORDS	WHAT TO SOLVE
when is the body at rest?	$v(t) = 0$
when does it change direction?	v changes sign
when is the speed greatest?	$a(t) = 0$
how far has it travelled?	total distance — count each direction separately

DISPLACEMENT IS NOT DISTANCE

A body that goes out 5 m and comes back has displacement 0 and distance 10 m. If v changes sign inside the interval, the journey must be split at that instant.

Connected rates

When two quantities are linked by a formula, their rates are linked by the chain rule:

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

Example. Water is poured into a cone at $20 \text{ cm}^3/s$. The cone has height 30 and top radius 15 cm. How fast is the depth rising when $h = 10$?

$$r = \frac{h}{2} \Rightarrow V = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h = \frac{\pi h^3}{12}$$

$$\frac{dV}{dh} = \frac{\pi h^2}{4} = 25\pi \Rightarrow \frac{dh}{dt} = \frac{20}{25\pi} \approx 0.255 \text{ cm/s}$$

ALWAYS REDUCE TO ONE VARIABLE FIRST

The relation $r = \frac{h}{2}$ comes from similar triangles and must be used **before** differentiating.

Differentiating a two-variable formula is the standard error here.

Cost and revenue

In economics the derivative of a total is called a **marginal** quantity: the approximate cost of producing **one more** item.

$$MC(q) = C'(q) \quad MR(q) = R'(q) \quad \text{profit is greatest when } MC = MR$$

The last statement is the optimisation of Lesson 23–25 in economic dress: profit $P = R - C$ is stationary when $P' = R' - C' = 0$.

Example. $C(q) = 0.02q^2 + 8q + 500$ so'm. The marginal cost at $q = 100$ is $C'(100) = 0.04 \times 100 + 8 = 12$ so'm — the 101st item costs about 12 so'm to make, while the average cost of the first hundred is $\frac{1500}{100} = 15$ so'm.

02 Worked examples

EXAMPLE 1 $s = t^3 - 6t^2 + 9t$ metres. Find when the body is at rest and the total distance in the first 4 s.

① $v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$

Rest at $t = 1, 3$.

② $s(0) = 0, s(1) = 4, s(3) = 0, s(4) = 4$

③ Distances: $4 + 4 + 4$.

Direction changes twice.

ANSWER

Rest at $t = 1$ and $t = 3$; distance 12 m

EXAMPLE 2 A ladder 5 m long slides down a wall. Its foot moves out at 0.4 m/s. How fast is the top falling when the foot is 3 m out?

① $x^2 + y^2 = 25$

Differentiate with respect to t .

② $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$

③ $x = 3 \Rightarrow y = 4$

④ $3(0.4) + 4 \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -0.3$

ANSWER

0.3 m/s downwards

EXAMPLE 3 $C(q) = 0.05q^2 + 12q + 800$. Find the marginal cost at $q = 60$ and the average cost.

① $C'(q) = 0.1q + 12$

② $C'(60) = 18$ so'm per item.

③ $C(60) = 180 + 720 + 800 = 1700$

④ Average $\frac{1700}{60} \approx 28.3$ so'm.

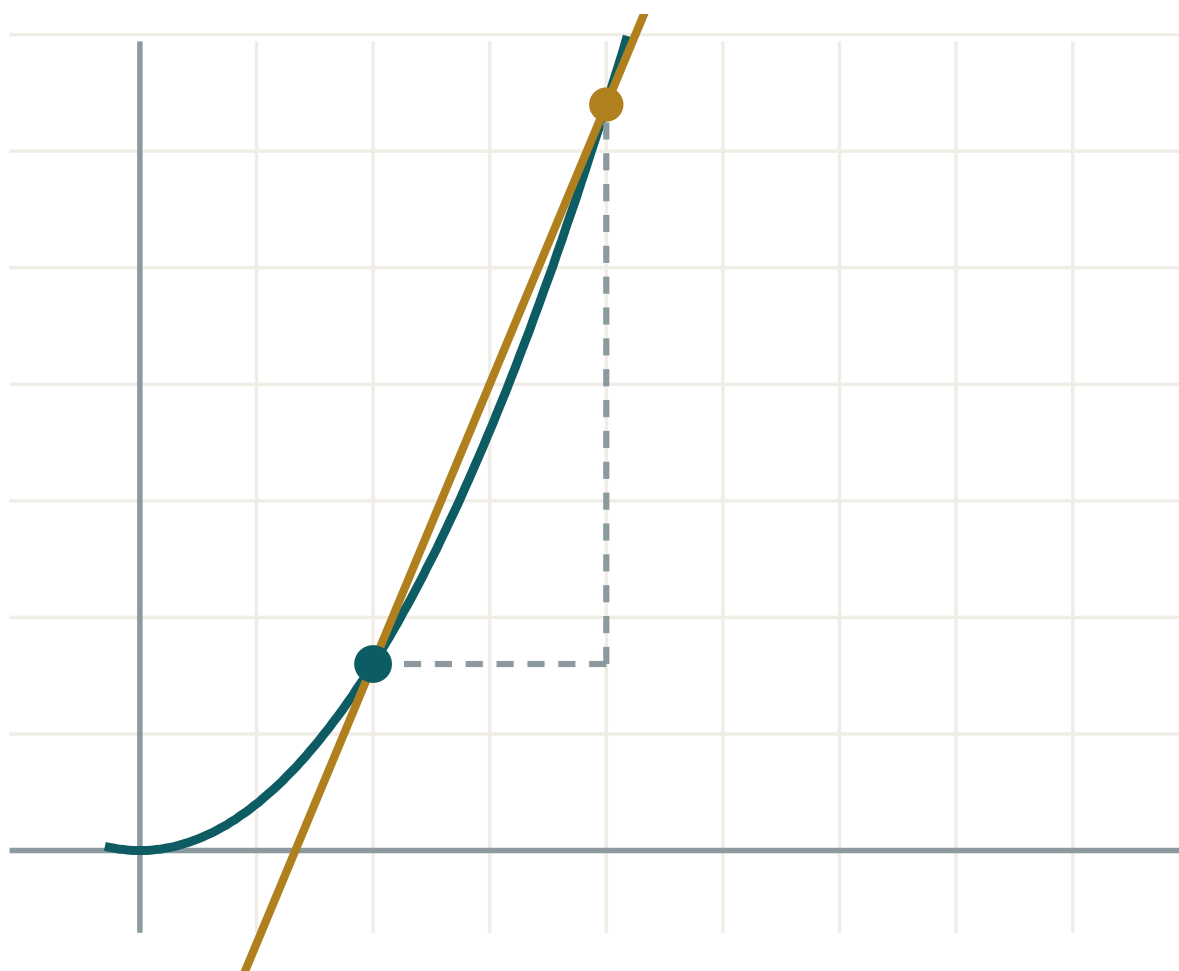
ANSWER

Marginal 18, average ≈ 28.3 so'm

03 Interactive model

Give every answer with its units, out loud, before writing it down.

A rate of change


 x 2

 h 2

 $\Delta y / \Delta x$ 2.4

 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Mathematical model	Matematik model	Математическая модель
Rate of change	O'zgarish tezligi	Скорость изменения
Displacement	Ko'chish	Перемещение
Velocity	Tezlik	Скорость
Acceleration	Tezlanish	Ускорение
Connected rates	Bog'liq tezliklar	Связанные скорости
Marginal cost	Marjinal xarajat	Предельные издержки
Marginal revenue	Marjinal daromad	Предельный доход
Units of a rate	Tezlik birligi	Единицы скорости
Instant of rest	To'xtash payti	Момент остановки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

If s is in metres and t in seconds, s' is in:

$$m$$

$$m/s$$

$$m/s^2$$

$$s/m$$

A body is at rest when:

$$s = 0$$

$$v = 0$$

$$a = 0$$

$$s'' = 0$$

Marginal cost is:

$$C(q)$$

$$C'(q)$$

$$\frac{C(q)}{q}$$

$$C''(q)$$

Connected rates use:

the product rule

the chain rule

the quotient rule

first principles

Profit is greatest when:

$$MC = 0$$

$$MR = 0$$

$$MC = MR$$

$$C = R$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $s = 5t^2$; find $v(3)$

ANSWER 30 m/s

2. $s = t^3$; find $a(2)$

ANSWER 12 m/s²

3. $s = 4t - t^2$; when is the body at rest?

ANSWER $t = 2$

4. Units of $\frac{dV}{dt}$ for volume in cm³

ANSWER cm³/s

5. $C(q) = 3q + 50$; marginal cost

ANSWER 3

6. $V = \frac{4}{3}\pi r^3$; find $\frac{dV}{dr}$

ANSWER $4\pi r^2$

7. $A = \pi r^2$; find $\frac{dA}{dr}$ at $r = 3$

ANSWER 6π

Medium · the standard question

7 PROBLEMS

1. $s = t^3 - 6t^2 + 9t$; when at rest?

ANSWER $t = 1, 3$

2. Same; total distance in the first 4 s

ANSWER 12 m

3. Balloon: $\frac{dr}{dt} = 2$ cm/s; find $\frac{dV}{dt}$ at $r = 5$

ANSWER 200π cm³/s

4. Circle: $\frac{dr}{dt} = 0.5$; find $\frac{dA}{dt}$ at $r = 8$

ANSWER 8π

5. Ladder 5 m, foot at 0.4 m/s; top's speed when foot is 3 m out

ANSWER 0.3 m/s

6. $C(q) = 0.05q^2 + 12q + 800$; marginal cost at $q = 60$

ANSWER 18

7. $s = 20t - 5t^2$; greatest height and when

ANSWER 20 m at $t = 2$

Hard · stretch and reason

7 PROBLEMS

1. Cone, height 30, radius 15, filled at $20 \text{ cm}^3/\text{s}$. $\frac{dh}{dt}$ at $h = 10$

ANSWER $\approx 0.255 \text{ cm/s}$

2. Same cone at $h = 20$

ANSWER $\approx 0.064 \text{ cm/s}$

3. $s = t^4 - 8t^2$; find all instants of rest

ANSWER $t = 0, \pm 2$

4. A cube's volume grows at $12 \text{ cm}^3/\text{s}$. Find $\frac{da}{dt}$ when $a = 4$

ANSWER 0.25 cm/s

5. Same cube: find $\frac{dS}{dt}$ at that moment

ANSWER $12 \text{ cm}^2/\text{s}$

6. $R(q) = 60q - 0.5q^2$, $C(q) = 10q + 200$. Maximise the profit

ANSWER $q = 50$, profit 1050

7. A trough 10 m long, cross-section an isosceles triangle 2 m wide and 1 m deep, fills at $0.5 \text{ m}^3/\text{min}$. Find $\frac{dh}{dt}$ at $h = 0.5$

ANSWER 0.05 m/min

07 Homework

Homework — 6 tasks

Every answer needs its units and a sentence in the words of the question.

- $s = 2t^3 - 15t^2 + 24t$. Find when the body is at rest, and the total distance in the first 5 s.
- A spherical balloon is inflated at $30 \text{ cm}^3/\text{s}$. Find how fast the radius grows when $r = 5 \text{ cm}$.

3. A ladder 13 m long slides; the foot moves at 0.5 m/s. Find the top's speed when the foot is 5 m out.
4. $C(q) = 0.1q^2 + 20q + 1000$. Find the marginal cost at $q = 40$ and the average cost.
5. $R(q) = 100q - q^2$ and $C(q) = 20q + 300$. Find the profit-maximising q .
6. A conical tank of height 12 m and top radius 6 m is filled at 3 m³/min. Find $\frac{dh}{dt}$ when $h = 4$ m.

Control work 3, and work on the mistakes

Approximation and modelling in one paper — the two topics where a missing unit costs a mark.

LESSON 35–36

2 × 40 MIN

ALGEBRA 11, NAZORAT ISHI 3

P1 · CHAPTER 9 REVIEW

LESSON CLOCK

3

35'

9'

22'

11'

Instructions	3 min
The paper	35 min
Self-mark	9 min
Rewrite	22 min
Units drill	11 min

LEARNING OBJECTIVES

- Apply linear approximation and connected rates under time.
- Give every answer with its units and in context.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 169–172
Cambridge	Pure Mathematics 1, pp. 191–194
Workbook	Control paper C

01

Explanation

The paper — 25 marks, 40 minutes

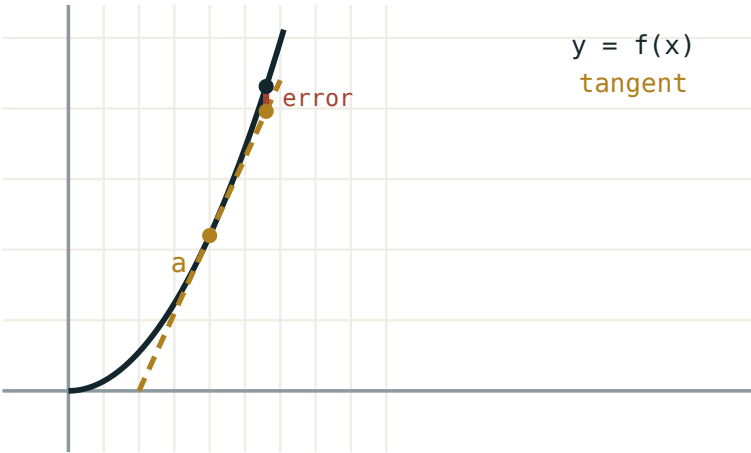
Q	TASK	MARKS	FROM
1	Estimate $\sqrt{37}$ and 2.02^4 by linear approximation	5	L28–30
2	A cube’s side is 12 ± 0.03 cm. Find the volume and its percentage error	4	L28–30
3	$s = t^3 - 9t^2 + 24t$: find when the body is at rest and the distance in the first 5 s	6	L31–34
4	A balloon inflates at $40 \text{ cm}^3/\text{s}$. Find $\frac{dr}{dt}$ when $r = 4$	5	L31–34
5	$C(q) = 0.04q^2 + 15q + 600$: marginal and average cost at $q = 50$	5	L31–34

UNITS ARE MARKED

Q3, Q4 and Q5 each carry one mark for the units alone. A numerically perfect answer with no units loses three of the twenty-five.

The three errors this topic produces

ERROR	LOOKS LIKE	KIND
units omitted	“the answer is 0.4”	careless
two variables kept	$\text{differentiating } V = \frac{1}{3}\pi r^2 h \text{ in } t \text{ without eliminating } r$	method
distance read as displacement	ignoring the change of direction	method
wrong a chosen	estimating $\sqrt{37}$ from $a = 25$	knowledge



near a the tangent and the curve agree

Question 1 in one picture — and the reason the choice of a matters.

The units drill

Ten rates on the board. For each, the class calls out only the **units**, in three seconds:

RATE	UNITS
$\frac{ds}{dt}$, s in km, t in hours	km/h
$\frac{dV}{dt}$, V in litres	<i>litres per second</i>
$\frac{dC}{dq}$, cost in so'm	so'm per item
$\frac{dA}{dr}$, A in cm^2	cm
$\frac{dP}{dt}$, P a population	people per year

ROW 4 IS THE INTERESTING ONE

$\frac{dA}{dr} = 2\pi r$ has units $cm^2/cm = cm$. A rate need not be “per second”: it is whatever the two quantities make it.

02

Worked examples

EXAMPLE 1 Model answer, Q1: estimate $\sqrt{37}$.

1

$a = 36, h = 1$
Nearest convenient square.

2

$f(36) = 6, f'(36) = \frac{1}{12}$

3

$6 + 1 \times \frac{1}{12} \approx 6.083$
True 6.0828.

ANSWER

≈ 6.083

EXAMPLE 2 Model answer, Q4: balloon at 40 cm³/s, find $\frac{dr}{dt}$ at $r = 4$.

$$\textcircled{1} \quad V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2 = 64\pi$$

$$\textcircled{2} \quad \frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$\textcircled{3} \quad 40 = 64\pi \cdot \frac{dr}{dt}$$

$$\textcircled{4} \quad \frac{dr}{dt} = \frac{40}{64\pi} \approx 0.199$$

ANSWER

≈ 0.20 cm/s

EXAMPLE 3 A learner wrote “the top of the ladder falls at 0.3”. What is missing?

$\textcircled{1}$ The number is right.

$\textcircled{2}$ No units, and no direction.
A careless error.

$\textcircled{3}$ Correct: 0.3 m/s downwards.

ANSWER

The units and the direction

03 Interactive model

Run the units drill before the rewrite, not after.

Units and estimates

Best a for estimating $\sqrt{37}$:

$\sqrt{37} \approx$

Units of $\frac{dA}{dt}$ for A in cm^2 :

Balloon at $40 \text{ cm}^3/\text{s}$, $r = 4$: $\frac{dr}{dt} \approx$

$s = t^3 - 9t^2 + 24t$: rest at

A 0.25% error in a cube's side gives what error in the volume?

04 Terminology

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ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Linear approximation	Chiziqli yaqinlashish	Линейное приближение
Connected rates	Bog'liq tezliklar	Связанные скорости
Units	O'lchov birliklari	Единицы измерения
Interpretation	Talqin	Интерпретация
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A rate's units come from:

the numerator only

both quantities

the context

nothing

Before differentiating a connected-rate formula you must:

square it

reduce it to one variable

integrate

nothing

Distance differs from displacement when:

never

the velocity changes sign

the acceleration is zero

always

Choosing a poor a in an estimate is:

careless

a method error

a knowledge gap

fine

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Estimate $\sqrt{37}$

ANSWER ≈ 6.083

2. Estimate 2.02^4

ANSWER ≈ 16.64

3. Units of $\frac{ds}{dt}$ for km and hours

ANSWER km/h

4. $s = t^2 - 4t$; when at rest?

ANSWER $t = 2$

5. $C(q) = 5q + 90$; marginal cost

ANSWER 5

6. $V = \frac{4}{3}\pi r^3$; $\frac{dV}{dr}$ at $r = 4$

ANSWER 64π

7. A 1% error in the side gives what error in the area?

ANSWER $\approx 2\%$

Medium · the standard question

7 PROBLEMS

1. Cube side 12 ± 0.03 ; percentage error in the volume

ANSWER 0.75%

2. $s = t^3 - 9t^2 + 24t$; rest at

ANSWER $t = 2, 4$

3. Same; distance in the first 5 s

ANSWER 28 m

4. Balloon at $40 \text{ cm}^3/\text{s}$; $\frac{dr}{dt}$ at $r = 4$

ANSWER $\approx 0.20 \text{ cm/s}$

5. $C(q) = 0.04q^2 + 15q + 600$; marginal cost at $q = 50$

ANSWER 19

6. Same; average cost at $q = 50$

ANSWER $\approx 32 \text{ so'm}$

7. Estimate $\frac{1}{5.02}$

ANSWER ≈ 0.1992

Hard · stretch and reason

7 PROBLEMS

1. Estimate $\sqrt[3]{27.5}$

ANSWER ≈ 3.0185

2. A cone of height 12, radius 6, fills at $4 \text{ m}^3/\text{min}$. $\frac{dh}{dt}$ at $h = 6$

ANSWER $\frac{4}{9\pi} \approx 0.141 \text{ m/min}$

3. $s = t^4 - 4t^3$; find every instant of rest

ANSWER $t = 0, 3$

4. A square's area grows at $8 \text{ cm}^2/\text{s}$. Find $\frac{da}{dt}$ at $a = 4$

ANSWER 1 cm/s

5. A ladder 10 m long; the top falls at 0.6 m/s . Find the foot's speed when the top is 6 m up

ANSWER 0.45 m/s

6. $R(q) = 80q - 0.4q^2$, $C(q) = 20q + 500$. Maximise the profit

ANSWER $q = 75$, profit 1750

7. Show that for $y = x^n$ the percentage error multiplies by n

ANSWER $\frac{\Delta y}{y} = n \frac{\Delta x}{x}$

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. Units on every line.

1. Rewrite in full every question that lost a mark, with units on every answer.
2. Five problems from the section your knowledge column was heaviest in.
3. Estimate $\sqrt{50}$ and 3.01^5 .

4. A cone of height 20 and top radius 10 fills at $5 \text{ cm}^3/\text{s}$. Find $\frac{dh}{dt}$ when $h = 8$.

The antiderivative and the indefinite integral

Differentiation run backwards — and the constant that makes the answer a whole family of curves.

LESSON 37–40

4 × 40 MIN

ALGEBRA 11, §2.3

P1 · 10.1

LESSON CLOCK

14

21'

23'

27'

41'

34'

The definition	14 min
Why the constant	21 min
Notation	23 min
Particular antiderivatives	27 min
Practice	41 min
Homework and consolidation	34 min

LEARNING OBJECTIVES

- State the definition of an antiderivative and verify one by differentiating.
- Explain why the antiderivative is determined only up to a constant.
- Write the indefinite integral with its constant of integration.
- Find the particular antiderivative through a given point.

TEXTBOOK

National	pp. 173–192
Cambridge	Pure Mathematics 1, pp. 196–202
Workbook	P1 Exercise 10A

01

Explanation

The definition

ANTIDERIVATIVE

F is an **antiderivative** of f on an interval when $F'(x) = f(x)$ for every x in it.

So the question “what is the antiderivative of $2x$?” asks: what did I differentiate to get $2x$? The answer x^2 is checked by differentiating it back.

$F(X)$	AN ANTIDERIVATIVE	CHECK
$2x$	x^2	$(x^2)' = 2x \checkmark$
$3x^2$	x^3	$(x^3)' = 3x^2 \checkmark$
1	x	$(x)' = 1 \checkmark$
0	7	$(7)' = 0 \checkmark$

ALWAYS CHECK BY DIFFERENTIATING

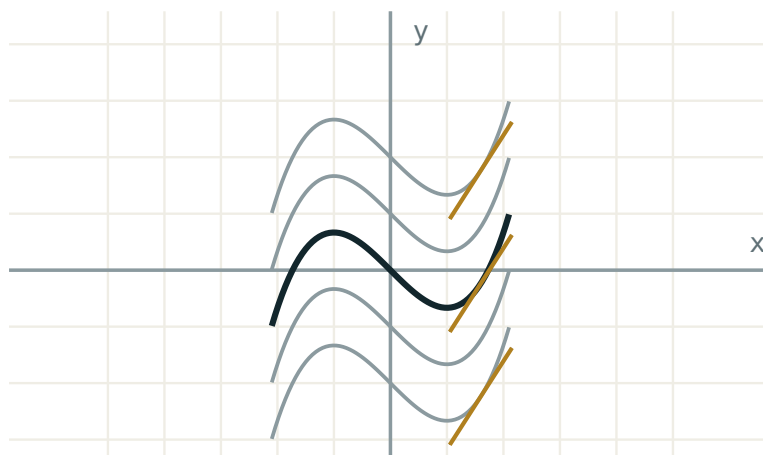
Integration has no algorithm that always works, but its answer can always be verified in one line. That check costs five seconds and catches almost every error.

Why there is a constant

The last row is the clue: 7 is an antiderivative of 0 , and so is -3 , and so is any constant. So if F works, so does $F + C$ — differentiating kills the constant.

$$(F + C)' = F' + 0 = f$$

Theorem. If F and G are both antiderivatives of f on an interval, then $F - G$ is constant. So the family $F + C$ is **all** of them.



every member has the same derivative — only C differs

*One shape, shifted. At any x the tangents are parallel
— that is what equal derivatives means.*

READ THE PICTURE

Every member of the family has the same gradient at each x . Knowing the derivative fixes the **shape** of the curve but not its **height**. One extra fact — a point on the curve — fixes the height.

The notation

$$\int f(x) dx = F(x) + C \text{ where } F'(x) = f(x)$$

Read “the integral of f of x , dee x ”. The $\int$ sign is an elongated S, for *summa*; the dx says which letter is the variable. The $+ C$ is part of the answer, not decoration.

+ C IS WORTH A MARK, EVERY TIME

An indefinite integral without its constant is an incomplete answer. It is the single most frequently lost mark in the whole of integration.

The particular antiderivative

An initial condition picks one member of the family:

Example. Find F with $F'(x) = 3x^2$ and $F(2) = 11$.

$$F(x) = x^3 + C \Rightarrow 8 + C = 11 \Rightarrow C = 3 \Rightarrow F(x) = x^3 + 3$$

In physics this is how a motion is reconstructed: the acceleration gives the velocity up to a constant, which the initial velocity supplies; the velocity gives the displacement up to another constant, which the initial position supplies.

$$a(t) = -10 \Rightarrow v(t) = -10t + v_0 \Rightarrow s(t) = -5t^2 + v_0t + s_0$$

Every formula of Grade-9 kinematics is an antiderivative, taken twice.

02 Worked examples**EXAMPLE 1** Find $\int 4x^3 dx$.

① What differentiates to $4x^3$?

② $(x^4)' = 4x^3$

③ Add the constant.

ANSWER

$$x^4 + C$$

EXAMPLE 2 Find the antiderivative of $f(x) = 6x^2 - 4x$ passing through $(1, 5)$.

① $F(x) = 2x^3 - 2x^2 + C$

Check: $F' = 6x^2 - 4x$ ✓

② $F(1) = 2 - 2 + C = C$

③ $C = 5$

ANSWER

$$F(x) = 2x^3 - 2x^2 + 5$$

EXAMPLE 3 A stone is dropped from 45 m. With $a = -10 \text{ m/s}^2$, when does it land?

① $v = -10t + 0$
Dropped, so $v_0 = 0$.

② $s = -5t^2 + 45$
 $s_0 = 45$.

③ $-5t^2 + 45 = 0 \Rightarrow t^2 = 9$

④ $t = 3 \text{ s}$.

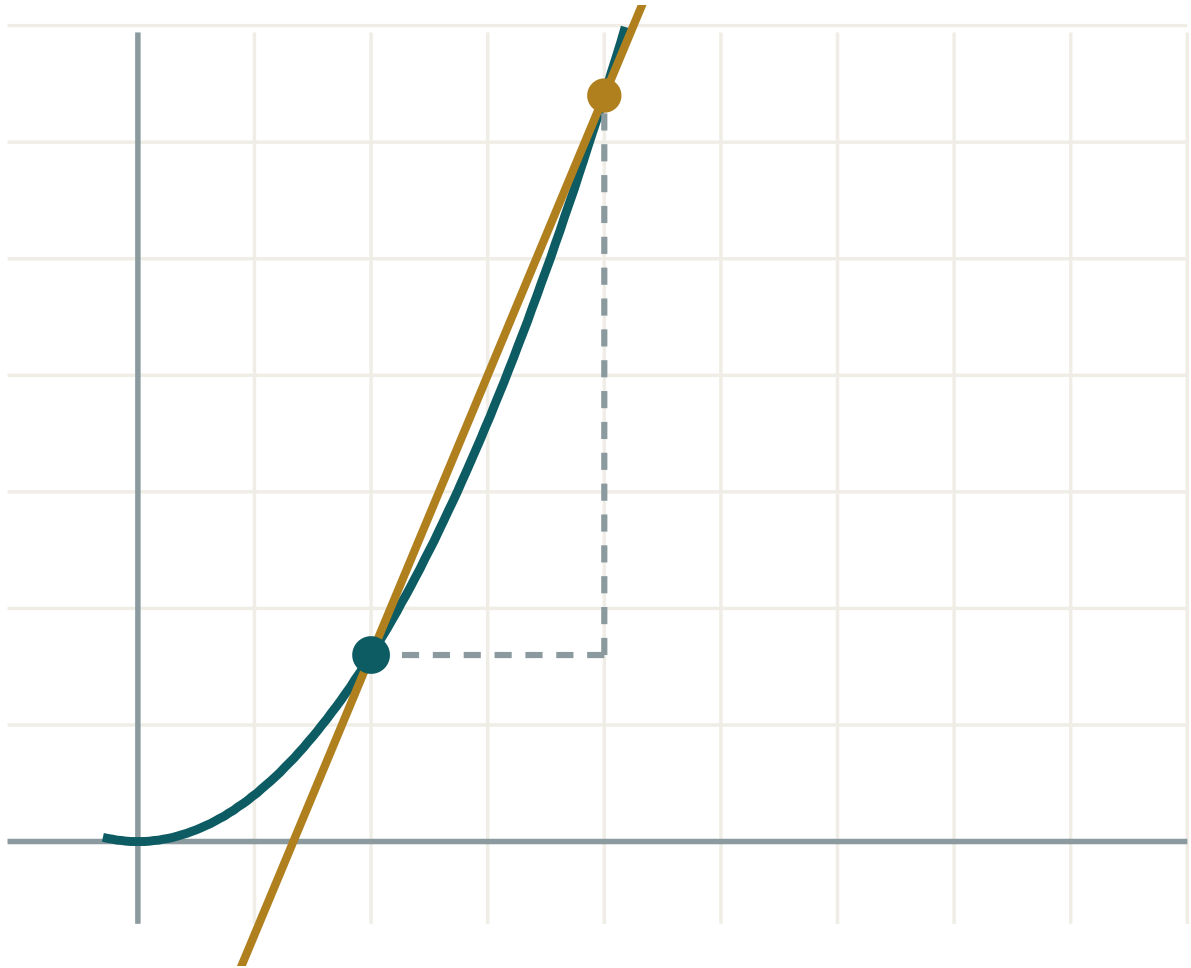
ANSWER

$t = 3 \text{ s}$

03 Interactive model

Differentiate every answer on the board before moving on.

Forwards and backwards



x 2

h 2

 $\Delta y / \Delta x$ 2.4 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Antiderivative	Boshlang'ich funksiya	Первообразная
Indefinite integral	Aniqmas integral	Неопределённый интеграл
Constant of integration	Integrallash doimiysi	Постоянная интегрирования
Integrand	Integral ostidagi funksiya	Подынтегральная функция
Family of curves	Egri chiziqlar oilasi	Семейство кривых
Initial condition	Boshlang'ich shart	Начальное условие
Particular solution	Xususiy yechim	Частное решение
Integration	Integrallash	Интегрирование
Inverse operation	Teskari amal	Обратная операция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

F is an antiderivative of f when:

$$F = f$$

$$F' = f$$

$$f' = F$$

$$F + f = 0$$

$\int 2x \, dx$ is:

$$x^2$$

$$x^2 + C$$

$$2$$

$$2x^2 + C$$

Two antiderivatives of the same function differ by:

nothing

a constant

a linear term

anything

$\int 0 \, dx$ is:

To fix C you need:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int 2x \, dx$

ANSWER $x^2 + C$

2. $\int 3x^2 \, dx$

ANSWER $x^3 + C$

3. $\int 1 \, dx$

ANSWER $x + C$

4. $\int 5 \, dx$

ANSWER $5x + C$

5. $\int 4x^3 \, dx$

ANSWER $x^4 + C$

6. $\int 0 \, dx$

ANSWER C

7. Is $x^2 + 7$ an antiderivative of $2x$?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. $\int (6x^2 - 4x) dx$

ANSWER $2x^3 - 2x^2 + C$

2. $\int (x + 3) dx$

ANSWER $\frac{x^2}{2} + 3x + C$

3. $\int (2x^3 - x) dx$

ANSWER $\frac{x^4}{2} - \frac{x^2}{2} + C$

4. Antiderivative of $6x^2 - 4x$ through $(1, 5)$

ANSWER $2x^3 - 2x^2 + 5$

5. Antiderivative of $4x$ through $(2, 10)$

ANSWER $2x^2 + 2$

6. $a = -10, v_0 = 0, s_0 = 45$. Find $s(t)$

ANSWER $45 - 5t^2$

7. When does that stone land?

ANSWER $t = 3 \text{ s}$

Hard · stretch and reason

7 PROBLEMS

1. Find F with $F''(x) = 6x$, $F'(0) = 2$ and $F(0) = 1$

ANSWER $F(x) = x^3 + 2x + 1$

2. Antiderivative of $(2x + 1)^2$ — expand first

ANSWER $\frac{4x^3}{3} + 2x^2 + x + C$

3. A car decelerates at 4 m/s^2 from 20 m/s . How far before it stops?

ANSWER 50 m

4. Prove that two antiderivatives differ by a constant

ANSWER $(F - G)' = 0$ on an interval

5. Show no f has $f'(x) = 3x^2$ and $f(1) = f(-1)$

ANSWER $f = x^3 + C$ gives $1 + C = -1 + C$, which is false

6. Why must the interval be stated in the theorem?

ANSWER On a disconnected domain the difference can jump

7. $\int (x + \frac{1}{x})^2 dx$ — expand and integrate the powers

ANSWER $\frac{x^3}{3} + 2x - \frac{1}{x} + C$

07 Homework

Homework — 6 tasks

Differentiate every answer to check it, and never omit $+ C$.

- Find $\int 5x^4 dx$, $\int (3x^2 + 2x) dx$ and $\int (x - 4) dx$.
- Find the antiderivative of $8x^3 - 6x$ passing through $(1, 0)$.
- Find F with $F'(x) = 12x^2$ and $F(2) = 40$.
- A ball is thrown up at 25 m/s from ground level, with $a = -10$. Find $s(t)$ and the greatest height.

5. Explain in three sentences why the answer to an indefinite integral is a family of curves.
6. Show that $x^2 + 3$ and $x^2 - 8$ are both antiderivatives of $2x$, and state their difference.

The table of integrals and the rules of integration

Six lessons on the standard integrals, the two linearity rules, and the one substitution that handles every bracket you will meet this year.

LESSON 41–46

6 × 40 MIN

ALGEBRA 11, §2.4

P1 · 10.2–10.3

LESSON CLOCK

18

18

27'

36'

36'

57'

48'

The power rule	18 min
The exceptional case	18 min
Linearity	27 min
Rewriting first	36 min
The linear bracket	36 min
Practice and homework	57 min
Consolidation	48 min

LEARNING OBJECTIVES

- Use the power rule for integration, including negative and fractional indices.
- Apply the constant-multiple and sum rules.
- Integrate $(ax + b)^n$.
- Rewrite a quotient or a product into integrable form before integrating.

TEXTBOOK

National	pp. 193–220
Cambridge	Pure Mathematics 1, pp. 203–212
Workbook	P1 Exercise 10B, 10C

01

Explanation

The power rule

$$\int x^n dx = \frac{x^{(n+1)}}{n + 1} + C, n \neq -1$$

Raise the index by one, divide by the new index. It is the power rule for differentiation

read backwards, and the check is immediate: differentiating $\frac{x^{(n+1)}}{n + 1}$ gives

$$\frac{(n+1)x^n}{n + 1} = x^n.$$

$F(X)$	$\int F(X) DX$
x^5	$\frac{x^6}{6} + C$
x	$\frac{x^2}{2} + C$
1	$x + C$
$\sqrt{x} = x^{1/2}$	$\frac{2}{3}x^{3/2} + C$
$\frac{1}{x^2} = x^{-2}$	$-\frac{1}{x} + C$
$\frac{1}{\sqrt{x}} = x^{-1/2}$	$2\sqrt{x} + C$

The one exception

$N = -1$ BREAKS THE RULE

$\int \frac{1}{x} dx$ cannot be $\frac{x^0}{0}$ — the division is by zero. The answer is $\ln|x| + C$, which comes from the logarithm, not from the power rule. At this stage, recognise the case and quote the result.

Every other index, positive, negative or fractional, obeys the rule. The exception is a single value, and it is the one every examiner tests.

Linearity

$$\int k f(x) dx = k \int f(x) dx \quad \int [f(x) \pm g(x)] dx = \int f dx \pm \int g dx$$

So a polynomial is integrated term by term, and one constant covers the whole answer:

$$\int (6x^2 - 4x + 5) dx = 2x^3 - 2x^2 + 5x + C$$

THERE IS NO PRODUCT OR QUOTIENT RULE FOR INTEGRATION

$\int f \cdot g \, dx$ is **not** $\int f \cdot \int g$. Test it on $\int x \cdot x \, dx$: the truth is $\frac{x^3}{3}$, but $\frac{x^2}{2} \cdot \frac{x^2}{2} = \frac{x^4}{4}$. Products and quotients must first be **rewritten**.

Rewriting first

Almost every integral at this level is done by turning the integrand into a sum of powers, then applying the rule:

GIVEN	REWRITE AS	INTEGRAL
$\frac{x^3 + 2x}{x}$	$x^2 + 2$	$\frac{x^3}{3} + 2x + C$
$x(x + 3)$	$x^2 + 3x$	$\frac{x^3}{3} + \frac{3x^2}{2} + C$
$(2x - 1)^2$	$4x^2 - 4x + 1$	$\frac{4x^3}{3} - 2x^2 + x + C$
$\frac{1}{x^3}$	x^{-3}	$-\frac{1}{2x^2} + C$
$x\sqrt{x}$	$x^{3/2}$	$\frac{2}{5}x^{5/2} + C$

The linear bracket

The chain rule run backwards gives one more standard result — enough for every bracket whose inside is **linear**:

$$\int (ax + b)^n \, dx = \frac{(ax + b)^{n+1}}{a(n+1)} + C, \quad n \neq -1$$

The extra a in the denominator undoes the a the chain rule would produce. Check $\int$

$$(3x + 1)^4 \, dx = \frac{(3x + 1)^5}{5} + C \text{ by differentiating: } \frac{5(3x+1)^4 \cdot 3}{5} = (3x + 1)^4 \checkmark.$$

ONLY WHEN THE INSIDE IS LINEAR

$\int (x^2 + 1)^4 \, dx$ cannot be done this way — the correction factor would have to be $2x$, which is not a constant. Expand it, or leave it for the next course.

02 Worked examples

EXAMPLE 1 Find $\int (4x^3 - \frac{6}{x^2} + \sqrt{x}) dx$.

① Rewrite: $4x^3 - 6x^{-2} + x^{1/2}$.

② x^4

③ $+ \frac{6}{x}$
 $-6 \cdot \frac{x^{-1}}{-1}$.

④ $+ \frac{2}{3}x^{3/2} + C$

ANSWER

$$x^4 + \frac{6}{x} + \frac{2}{3}x\sqrt{x} + C$$

EXAMPLE 2 Find $\int \frac{x^2 - 3x}{x} dx$.

① Divide first: $x - 3$.
 No quotient rule exists.

② $\frac{x^2}{2} - 3x + C$

ANSWER

$$\frac{x^2}{2} - 3x + C$$

EXAMPLE 3 Find $\int (2x + 5)^3 dx$.

- ① Linear inside, so use the bracket rule.

$$a = 2, n = 3$$

②
$$\frac{(2x + 5)^4}{2 \times 4}$$

③
$$= \frac{(2x + 5)^4}{8} + C$$

Check by differentiating.

ANSWER

$$\frac{(2x + 5)^4}{8} + C$$

03 Interactive model

Differentiate every answer aloud before writing the next question.

Which rule, and what is the answer?

 $\int x^4 dx$:

$4x^3 + C$

$\frac{x^5}{5} + C$

$x^5 + C$

$\frac{x^3}{3} + C$

 $\int \frac{1}{x^2} dx$:

$\ln|x| + C$

$-\frac{1}{x} + C$

$\frac{1}{x} + C$

$-2x^{-3} + C$

 $\int \frac{1}{x} dx$:

$\frac{x^0}{0}$

$\ln|x| + C$

0

undefined

$\int (3x + 1)^4 dx$:

$$\frac{(3x+1)^5}{5} + C$$

$$\frac{(3x+1)^5}{15} + C$$

$$3(3x+1)^5 + C$$

$$(3x+1)^5 + C$$

$\int x \cdot x dx$:

$$\frac{x^4}{4} + C$$

$$\frac{x^3}{3} + C$$

$$x^2 + C$$

$$\frac{x^2}{2} + C$$

$\int \sqrt{x} dx$:

$$\frac{2}{3} x^{(3/2)} + C$$

$$\frac{1}{2\sqrt{x}} + C$$

$$\frac{3}{2} x^{(1/2)} + C$$

$$x^{(3/2)} + C$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Table of integrals	Integrallar jadvali	Таблица интегралов
Power rule for integration	Daraja integrallash qoidasi	Правило интегрирования степени
Constant multiple rule	O'zgarmas ko'paytuvchi qoidasi	Правило постоянного множителя
Sum rule	Yig'indi qoidasi	Правило суммы
Linear substitution	Chiziqli almashtirish	Линейная замена
Integrable form	Integrallanuvchi shakl	Интегрируемый вид
Reciprocal function	Teskari funksiya ($1/x$)	Обратная величина
Exceptional case	Istisno hol	Особый случай
Term by term	Hadma-had	Почленно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The power rule for integration fails when:

$\int \frac{1}{x} dx$ is:

Is there a product rule for integration?

$\int (ax + b)^n dx$ divides by:

$n + 1$

a

$a(n + 1)$

an

Before integrating $\frac{x^3 + x}{x}$ you should:

use a quotient rule

divide through

square it

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int x^3 dx$

ANSWER $\frac{x^4}{4} + C$

2. $\int x^6 dx$

ANSWER $\frac{x^7}{7} + C$

3. $\int 5 dx$

ANSWER $5x + C$

4. $\int 2x dx$

ANSWER $x^2 + C$

5. $\int (x + 1) dx$

ANSWER $\frac{x^2}{2} + x + C$

6. $\int x^{-2} dx$

ANSWER $-\frac{1}{x} + C$

7. $\int \sqrt{x} dx$

ANSWER $\frac{2}{3}x^{(3/2)} + C$

Medium · the standard question

7 PROBLEMS

1. $\int (3x^2 - 4x + 1) dx$

ANSWER $x^3 - 2x^2 + x + C$

2. $\int \frac{1}{x^3} dx$

ANSWER $-\frac{1}{2x^2} + C$

3. $\int \frac{1}{\sqrt{x}} dx$

ANSWER $2\sqrt{x} + C$

4. $\int x(x + 2) dx$

ANSWER $\frac{x^3}{3} + x^2 + C$

5. $\int \frac{x^3 + 2x}{x} dx$

ANSWER $\frac{x^3}{3} + 2x + C$

6. $\int (2x + 5)^3 dx$

ANSWER $\frac{(2x+5)^4}{8} + C$

7. $\int (1 - x)^5 dx$

ANSWER $-\frac{(1-x)^6}{6} + C$

Hard · stretch and reason

7 PROBLEMS

1. $\int (2x - 1)^2 dx$

ANSWER $\frac{4x^3}{3} - 2x^2 + x + C$

2. $\int x\sqrt{x} dx$

ANSWER $\frac{2}{5}x^{5/2} + C$

3. $\int (x + \frac{1}{x})^2 dx$

ANSWER $\frac{x^3}{3} + 2x - \frac{1}{x} + C$

4. $\int \frac{(x+1)^2}{x^2} dx$

ANSWER $x + 2\ln|x| - \frac{1}{x} + C$

5. $\int \frac{1}{(3x-2)^2} dx$

ANSWER $-\frac{1}{3(3x-2)} + C$

6. $\int \sqrt{4x+1} dx$

ANSWER $\frac{(4x+1)^{3/2}}{6} + C$

7. Why can $\int (x^2 + 1)^4 dx$ not use the bracket rule?

ANSWER The inside is not linear

07 Homework**Homework — 7 tasks**

Every answer differentiated back, and + C on every line.

1. Find $\int (5x^4 - 3x^2 + 2) dx$.

2. Find $\int (\sqrt{x} + \frac{1}{x^2}) dx$.
3. Find $\int \frac{x^4 - x}{x^2} dx$.
4. Find $\int (3x - 4)^5 dx$.
5. Find $\int (x + 2)(x - 3) dx$.
6. Find $\int \frac{1}{\sqrt{2x + 1}} dx$.
7. Explain in three sentences why there is no product rule for integration, with an example.

Control work 4, and the quarter review

The indefinite integral in one paper, and the map that joins approximation, modelling and integration.

LESSON 47–48

2 × 40 MIN

ALGEBRA 11, NAZORAT ISHI 4

P1 · CHAPTER 10 REVIEW

LESSON CLOCK

3

37'

9'

18'

9'

4'

Instructions	3 min
The paper	37 min
Answers	9 min
Rewrite	18 min
Concept map	9 min
Targets	4 min

LEARNING OBJECTIVES

- Apply every technique of Quarter II in one assessment.
- Rewrite an integrand into integrable form without prompting.
- Build a concept map of the quarter.
- Set a personal target for Quarter III.

TEXTBOOK

National	pp. 221–224
Cambridge	Pure Mathematics 1, pp. 213–216
Workbook	Control paper D

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Find $\int (6x^2 - \frac{4}{x^2} + 3\sqrt{x}) dx$	5	L41– 46
2	Find $\int \frac{x^3 - 2x}{x} dx$ and $\int (4x - 3)^5 dx$	5	L41– 46
3	Find F with $F'(x) = 3x^2 - 2x$ and $F(2) = 10$	4	L37– 40
4	Estimate $\sqrt[3]{63}$ and give the percentage error in a cube's volume from a 0.5% error in its side	5	L28– 30
5	$s = t^3 - 12t$: find when the body is at rest and its acceleration then	5	L31– 34
6	A cone of height 24 and top radius 8 fills at $6 \text{ cm}^3/\text{s}$. Find $\frac{dh}{dt}$ at $h = 12$	6	L31– 34

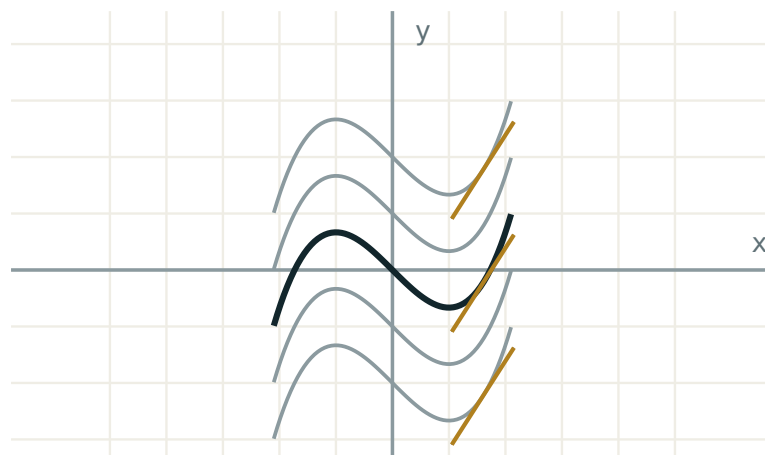
THREE MARKS ARE FOR $+C$ AND UNITS

Q1, Q2 and Q3 each lose a mark without the constant; Q5 and Q6 each lose one without units. Four of the thirty marks are notation.

The concept map

Six boxes, links as sentences:

- **derivative** → **linear approximation** — “near a the tangent is the curve”
- **derivative** → **rate of change** — “with units, in context”
- **rate** → **connected rates** — “the chain rule links two rates”
- **derivative** → **antiderivative** — “run the arrow backwards”
- **antiderivative** → $+C$ — “the derivative forgets the height”
- **rules of differentiation** → **rules of integration** — “power up, divide; linearity survives; product does not”



every member has the same derivative – only C differs

The picture behind two of the six boxes.

Looking forward

Quarter III opens with the definite integral and the Newton–Leibniz formula, which turns every antiderivative of this quarter into an area. Nothing new about integration itself is added first — what is added is a pair of limits and a subtraction.

ONE HABIT TO CARRY FORWARD

Rewrite before integrating. Roots into fractional indices, fractions into negative indices, products expanded, quotients divided through. The integral is then one line.

02 Worked examples

EXAMPLE 1 Model answer, Q1: $\int (6x^2 - \frac{4}{x^2} + 3\sqrt{x}) \, dx$.

① Rewrite: $6x^2 - 4x^{-2} + 3x^{1/2}$.

② $2x^3$

③ $+ \frac{4}{x}$
 $-4 \cdot \frac{x^{-1}}{-1}$.

④ $+ 2x^{3/2} + C$
 $3 \cdot \frac{2}{3}$.

ANSWER

$$2x^3 + \frac{4}{x} + 2x\sqrt{x} + C$$

EXAMPLE 2 Model answer, Q6: cone, height 24, top radius 8, filling at $6 \text{ cm}^3/\text{s}$.

$$\textcircled{1} \quad r = \frac{h}{3}$$

Similar triangles.

$$\textcircled{2} \quad V = \frac{1}{3} \pi \frac{h^2}{9} h = \frac{\pi h^3}{27}$$

$$\textcircled{3} \quad \frac{dV}{dh} = \frac{\pi h^2}{9} = 16\pi \text{ at } h = 12.$$

$$\textcircled{4} \quad \frac{dh}{dt} = \frac{6}{16\pi} \approx 0.119$$

ANSWER $\approx 0.12 \text{ cm/s}$ **EXAMPLE 3** Model answer, Q5: $s = t^3 - 12t$.

$$\textcircled{1} \quad v = 3t^2 - 12 = 3(t - 2)(t + 2)$$

$$\textcircled{2} \quad t = 2 \text{ (taking } t \geq 0\text{)}.$$

$$\textcircled{3} \quad a = 6t = 12$$

ANSWERRest at $t = 2 \text{ s}$; $a = 12 \text{ m/s}^2$ **03 Interactive model**

Work Q1 and Q6 on the board, rewriting the integrand before touching it.

The quarter in ten questions

Estimate $\sqrt{63}$:

7.94

7.9375

7.8

8.06

A 0.5% error in a side gives what error in the volume?

0.5%

1%

1.5%

3%

$s = t^3 - 12t$: rest at

$t = 2$

$t = 4$

$t = 12$

$t = 0$

The acceleration there:

6

12

24

0

$\int 6x^2 dx$:

$12x + C$

$2x^3 + C$

$6x^3 + C$

$3x^3 + C$

$\int \frac{4}{x^2} dx$:

$-\frac{4}{x} + C$

$\frac{4}{x} + C$

$4\ln|x| + C$

$-\frac{8}{x^3} + C$

$\int (4x - 3)^5 dx$:

$$\frac{(4x-3)^6}{6} + C$$

$$\frac{(4x-3)^6}{24} + C$$

$$4(4x-3)^6 + C$$

$$(4x-3)^6 + C$$

$F' = 3x^2 - 2x$, $F(2) = 10$ gives C :

$$0$$

$$6$$

$$10$$

$$-4$$

$\int \frac{1}{x} dx$:

$$\frac{x^0}{0}$$

$$\ln|x| + C$$

$$0$$

$$x + C$$

Cone height 24, radius 8, at $6 \text{ cm}^3/\text{s}$: $\frac{dh}{dt}$ at $h = 12$

0.12

0.24

1.2

0.02

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Integrable form	Integrallanuvchi shakl	Интегрируемый вид
Constant of integration	Integrallash doimiysi	Постоянная интегрирования
Initial condition	Boshlang'ich shart	Начальное условие
Self-assessment	O'z-o'zini baholash	Самооценка
Target	Maqsad	Цель
Revision	Takrorlash	Повторение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The commonest lost mark in integration is:

a sign

+ C

the index

the units

Before integrating a quotient you should:

use a quotient rule

divide through

square it

differentiate

Quarter III begins with:

probability

the definite integral

complex numbers

vectors

A connected-rate problem must first be reduced to:

- two variables
- one variable
- three variables
- a constant

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\int 6x^2 dx$

ANSWER $2x^3 + C$

2. $\int \frac{4}{x^2} dx$

ANSWER $-\frac{4}{x} + C$

3. $\int 3\sqrt{x} dx$

ANSWER $2x^{3/2} + C$

4. Estimate $\sqrt{63}$

ANSWER ≈ 7.9375

5. $s = t^3 - 12t$; rest at

ANSWER $t = 2$

6. Acceleration there

ANSWER 12

7. $\int (4x - 3)^5 dx$

ANSWER $\frac{(4x-3)^6}{24} + C$

Medium · the standard question

7 PROBLEMS

1. $\int (6x^2 - \frac{4}{x^2} + 3\sqrt{x}) dx$

ANSWER $2x^3 + \frac{4}{x} + 2x^{3/2} + C$

2. $\int \frac{x^3 - 2x}{x} dx$

ANSWER $\frac{x^3}{3} - 2x + C$

3. F with $F' = 3x^2 - 2x$, $F(2) = 10$

ANSWER $x^3 - x^2 + 6$

4. 0.5% error in a side; error in the volume

ANSWER 1.5%

5. Cone $24/8$ at $6 \text{ cm}^3/\text{s}$; $\frac{dh}{dt}$ at $h = 12$

ANSWER $\approx 0.12 \text{ cm/s}$

6. $\int (x + 3)(x - 1) dx$

ANSWER $\frac{x^3}{3} + x^2 - 3x + C$

7. Estimate 4.02^3

ANSWER ≈ 64.96

Hard · stretch and reason

7 PROBLEMS

1. $\int \frac{(x+2)^2}{x^2} dx$

ANSWER $x + 4\ln|x| - \frac{4}{x} + C$

2. $\int \sqrt{3x+2} dx$

ANSWER $\frac{2(3x+2)^{3/2}}{9} + C$

3. $\int \frac{1}{(2x-5)^3} dx$

ANSWER $-\frac{1}{4(2x-5)^2} + C$

4. Find F with $F'' = 12x$, $F'(1) = 4$, $F(1) = 5$

ANSWER $2x^3 - 2x + 5$

5. A trough fills at $0.6 \text{ m}^3/\text{min}$; find $\frac{dh}{dt}$ when the surface is 3 m^2 in area

ANSWER 0.2 m/min

6. A car decelerating at 5 m/s^2 from 30 m/s : stopping distance

ANSWER 90 m

7. Explain why $\int (x^2 + 1)^3 dx$ must be expanded

ANSWER The bracket rule needs a linear inside

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.

3. Find $\int (8x^3 - \frac{3}{x^2} + 4\sqrt{x}) dx$ and $\int (5x + 2)^4 dx$.
4. Write your target for Quarter III in one checkable sentence, and date it.